Name: _____ Candidate Number: _____

City of London School

ENTRANCE EXAMINATION 2008
MATHEMATICS

GROUP 1 TIME: 45 minutes

Answer as many questions as you can in the spaces provided.
Show all your working clearly.
Be careful not to spend too long on any one question.
You will need a ruler and pencil (or pen).

No calculators are allowed.
No spare paper will be provided.

1. (a) Gold ribbon costs **60p for one metre**. Tom has **£2.40**
How many metres of gold ribbon can he buy?

..... metres

(b) Blue ribbon costs **40p for one metre**. Nicola buys $\mathbf{3\frac{1}{2}}$ **metres**.
How much does this cost?

……..

2. Work out the following calculations.

a)

$$\begin{array}{r} 524 \\ +\,482 \\ \hline \end{array}$$

b)

$$\begin{array}{r} 956 \\ -\,492 \\ \hline \end{array}$$

c)

$$\begin{array}{r} 89 \\ \times\ 7 \\ \hline \end{array}$$

d)

$$9\overline{)648}$$

3. a) Tim, his mother and his grandmother all have their birthday on the same day.



Tim
12 years old



Tim's mother
37 years old



Tim's grandmother
70 years old

(i) When Tim was born, how old was **Tim's grandmother**?

..... years old

(ii) When **Tim's mother** is **60** years old, how old will **Tim** be?

..... years old

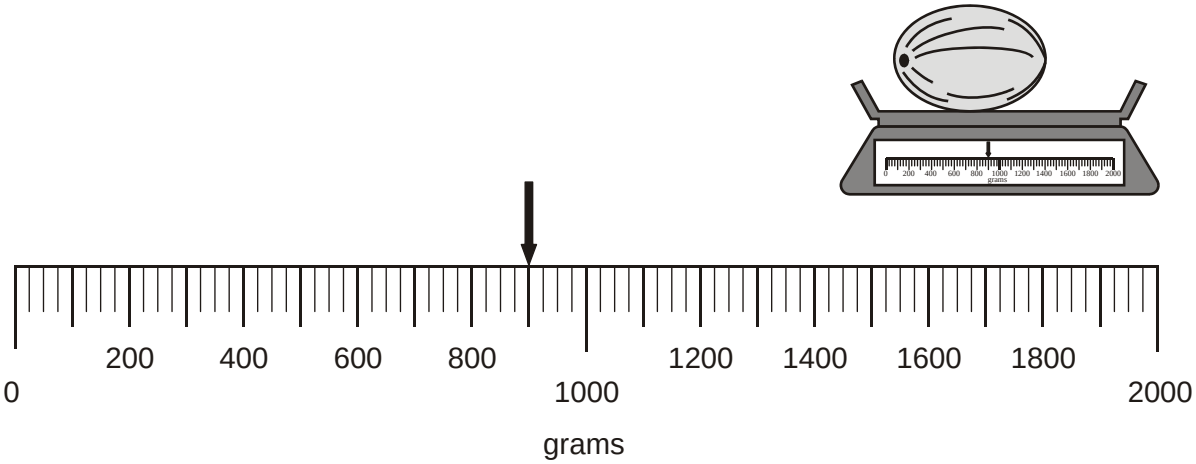
b) I am thinking of two numbers.

When I **add** my numbers, the answer is **11**
When I **multiply** my numbers, the answer is **24**

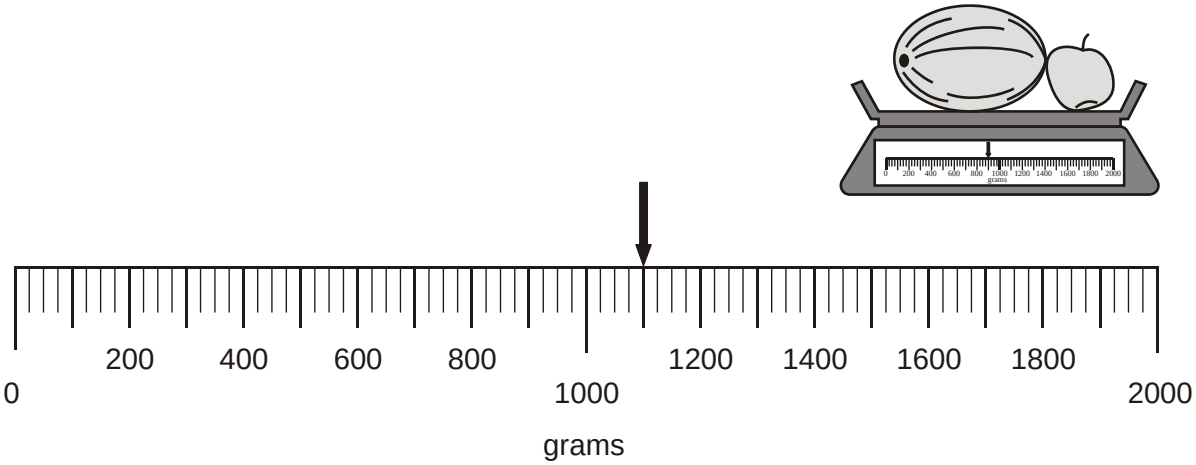
What are my numbers?

…… and ……

4. (a) I weigh a melon.



Then I weigh an apple and the melon.



Write the missing numbers in the sentences below.

The **melon** weighs ..... grams.

The **apple** weighs ..... grams.

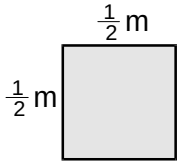
(b) **How many grams** are in one **kilogram**?

Put a ring round the correct number below.

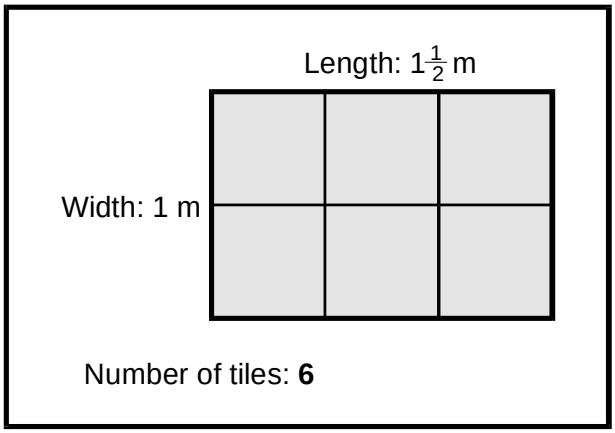
1 10 100 1000 10 000

5. (a) A shop sells square carpet tiles.

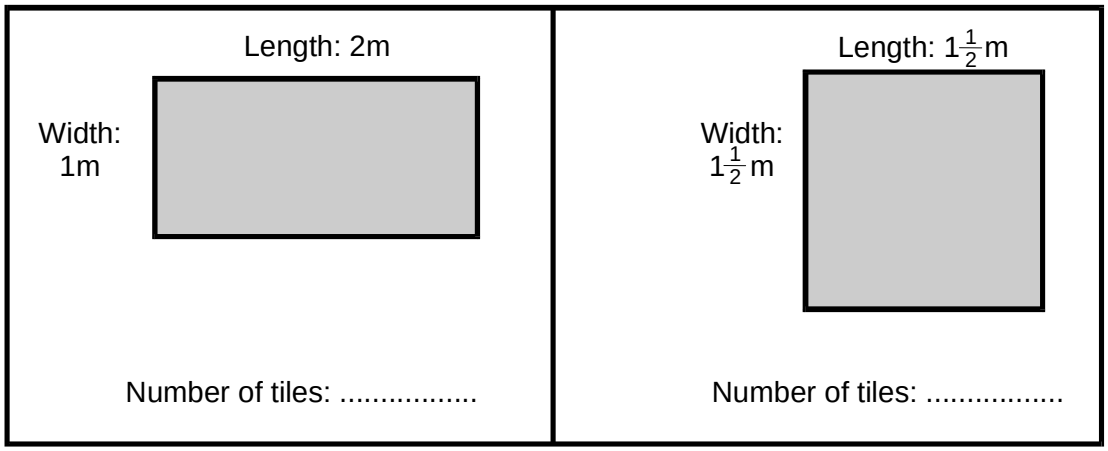
The edge of a carpet tile is $\frac{1}{2}$ m long.



6 carpet tiles can fit together on a floor like this:



Write how many carpet tiles fit on each of the two floors shown below.
You can draw the tiles if you want to.



The table shows how many carpet tiles you need for some bigger floors.

		length of room				
		3m	**3 $\frac{1}{2}$ m**	**4m**	**4 $\frac{1}{2}$ m**	**5m**
width of room	**3m**	36	42	48	54	60
	3 $\frac{1}{2}$ m	42	49	56	63	70
	4m	48	56	64	72	80
	4 $\frac{1}{2}$ m	54	63	72	81	90
	5m	60	70	80	90	100

(b) Doctor Gee wants to carpet a floor that is **4m** by **3½ m.**

i) How many carpet tiles does she need?

..... tiles

ii) The shop sells the carpet tiles only in **packs of 10.**
How many packs of 10 should Doctor Gee buy?

..... packs of 10

6. (a) A shopkeeper stacks tins.

In each layer there are the same number of tins.

How many tins are in each stack below?



....

(b) The shopkeeper stacks jars in layers.

Each layer looks like this.



How many jars will be in the stack when it is **5 layers** high?

.....

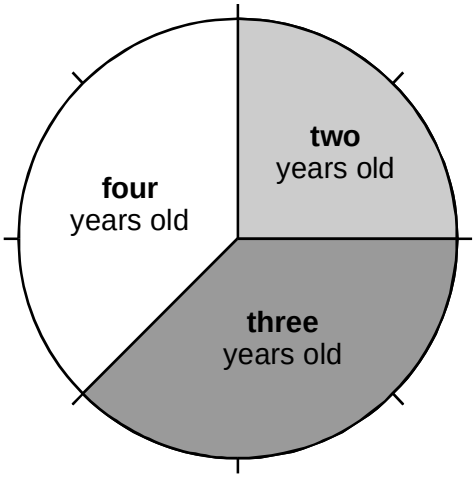
(c) The shopkeeper stacks **45 boxes** in layers.

In each layer there are **5** boxes.

How many layers high is the stack?

.....

7. The pie chart shows information about children who go to a nursery school.



Altogether, **80 children** go to the nursery school.

(a) **How many** of the 80 children are **two** years old?

.....

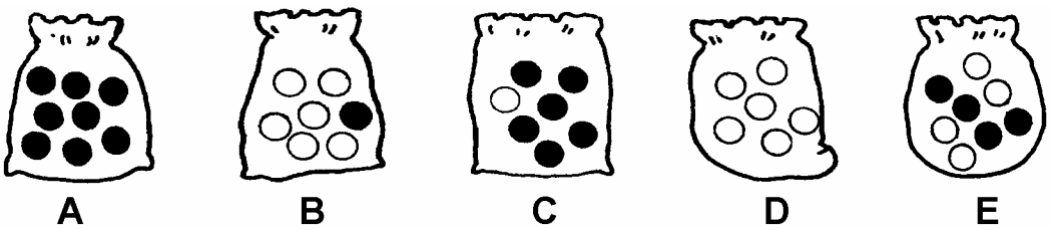
(b) **How many** of the 80 children are **four** years old?

.....

8. Bryn has some bags with some black beads and some white beads.

He is going to take a bead from each bag without looking.

(a) Match the pictures to the statements. The first is done for you.



It is **impossible** that Bryn will take a black bead from bag **…D…**

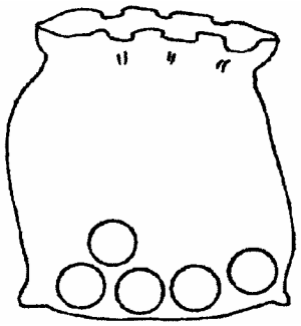
It is **unlikely** that Bryn will take a black bead from bag **……..**

It is **equally likely** that Bryn will take
a black bead or a white bead from bag **……..**

It is **likely** that Bryn will take a black bead from bag **……..**

It is **certain** that Bryn will take a black bead from bag **……..**

(b) Bryn has **5 white** beads in a bag.



He wants to make it **more likely** that he will take a **black** bead than a **white** bead out of the bag.

How many **black** beads should Bryn put into the bag?

.....

(c) There are **20** beads altogether in another bag.

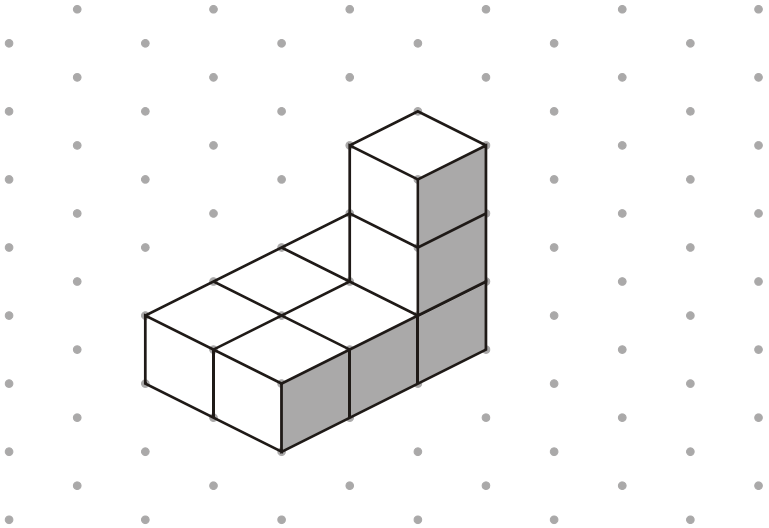
All the beads are either black or white.

It is **equally likely** that Bryn will take a black bead or a white bead from the bag.

How many black beads and how many white beads are there in the bag?

..... black beads and ..... white beads

9. Sam uses 8 cubes to make this shape.

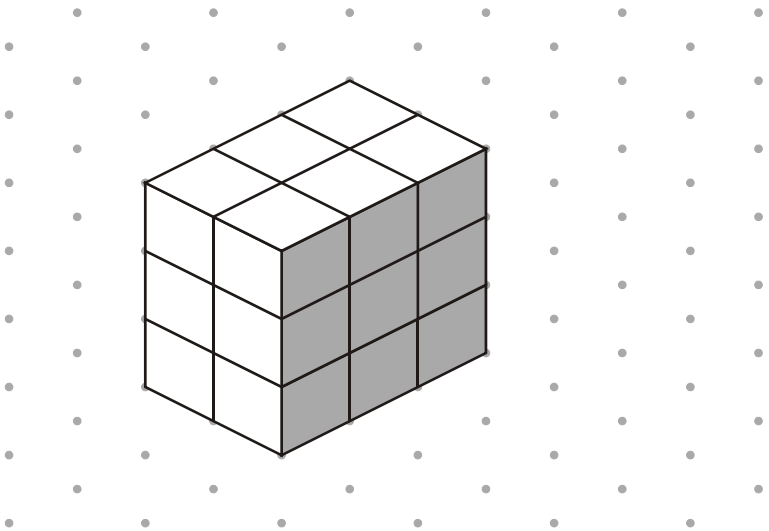


a) The volume of each cube is $2cm^3$.

What is the volume of the shape Sam made?

.....cm^3

b) He adds more cubes to make this cuboid.



How many **more** cubes does he add to make this cuboid?

.....

10. a) Write down three even numbers that add to twenty.

b) What is the next number in the sequence of square numbers?

One, four, nine, sixteen, ...

11. In each part, put a tick (✓) by the statement which is **most likely** to be **true**.

(a) 

The **kettle** in my kitchen holds **2 litres** of water.	
The **kettle** in my kitchen holds **20 litres** of water.	
The **kettle** in my kitchen holds **200 litres** of water.	
The **kettle** in my kitchen holds **2000 litres** of water	

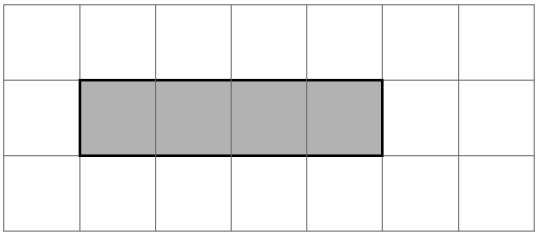
(b) 

The **door** to my room is **2mm** high.	
The **door** to my room is **2cm** high.	
The **door** to my room is **2m** high.	
The **door** to my room is **2km** high.	

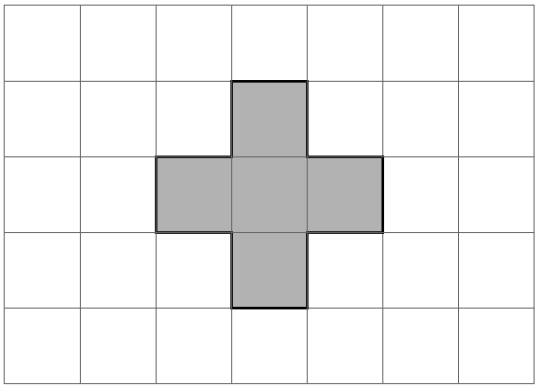
(c) 

The apple **in my bag weighs** 1 gram.	
The apple **in my bag weighs** 10 grams.	
The apple **in my bag weighs** 100 grams.	
The apple **in my bag weighs** 1000 grams.	

12. The shaded rectangle has an **area** of **4 cm^2** and a **perimeter** of 1**0 cm**.



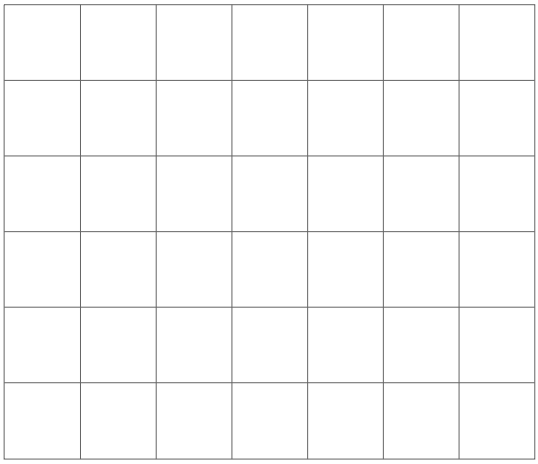
(a) Look at the cross-shape.



Fill in the gaps below. The cross-shape has

an **area** of ..... **cm^2** and a **perimeter** of ..... **cm.**

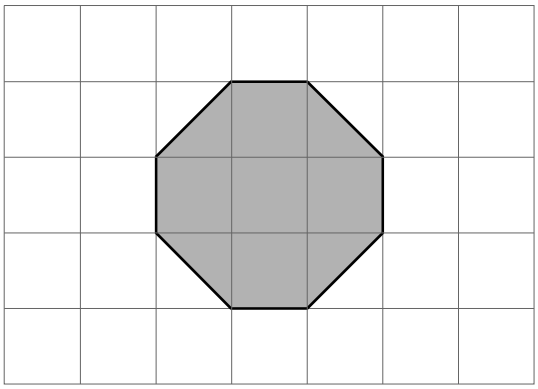
(b) Draw a shape with an **area** of **6 cm^2**



(c) What is the **perimeter** of your shape?

..... cm

(d) Look at the octagon.



What is the area of the octagon?

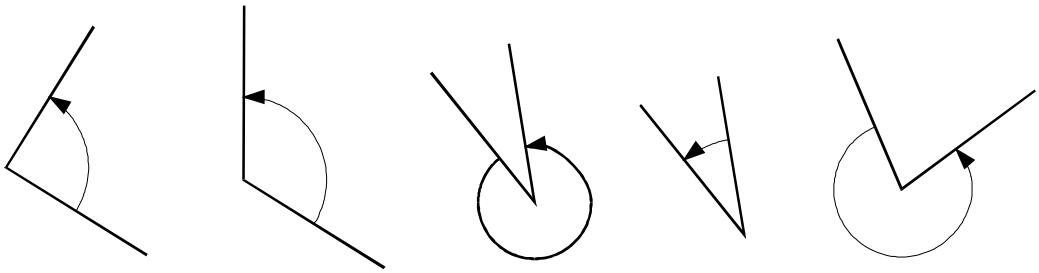
..... cm^2

(e) Explain how you know that the perimeter of the octagon is **more than 8cm**.

……

……

13. Look at these angles.



angle P angle Q angle R angle S angle T

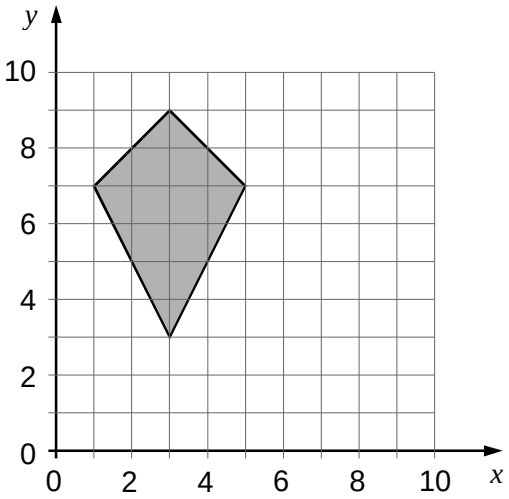
One of the angles measures **120°**

Write its letter.

.....

14.

Look at the shaded shape.



(a) **Two** statements below are correct.

Tick the correct statements.

The shape is a **quadrilateral.** ☐

The shape is a **trapezium**. ☐

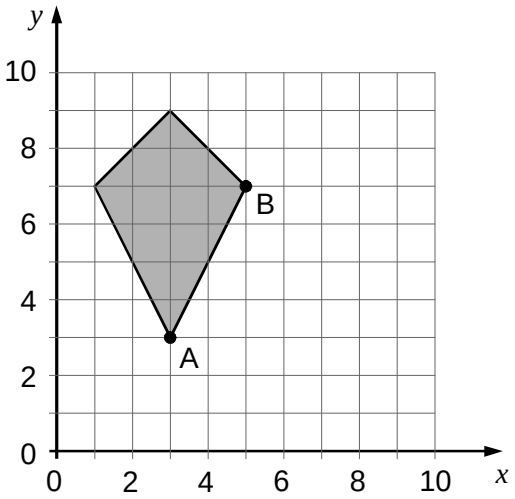
The shape is a **pentagon**. ☐

The shape is a **kite.** ☐

The shape is a **parallelogram**. ☐

(b) What are the co-ordinates of point B?

(,)

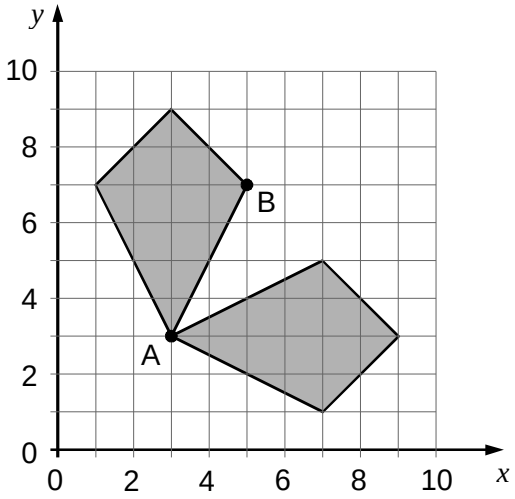


(c) The shape is **reflected** in a mirror line.

Point A stays in the same place.

Where is point **B** reflected to?

Put a cross on the grid to show the correct place.



15. (a) Peter's height is **0.9m**.

Lucy is **0.3m taller** than Peter.

What is Lucy's height?

.....m

(b) Lee's height is **1.45m.**

Misha is **0.3m shorter** than Lee.

What is Misha's height?

.....m

(c) Zita's height is **1.7m.**

What is Zita's height in **centimetres**?

.....cm

16. i) Write down a number that is a factor of both nine and twelve.

.....

ii) Work out the number of boys and girls in each class below.

(a) In class 8M, there are **27 pupils**.

There are **twice as many boys** as girls.

Number of boys	Number of girls
.....	.....

(b) In class 8K, there are **28 pupils**.

There are **two more boys** than girls.

Number of boys	Number of girls
.....	.....

17. Work out the missing numbers. In each part, you can use the first line to help you.

(a)

16	×	15	=	240
16	×	.....	=	480

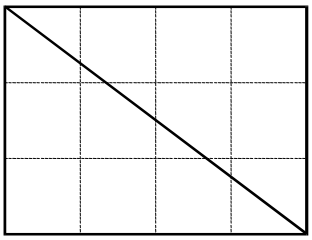
(b)

46	×	44	=	2024
46	×	22	=	.....

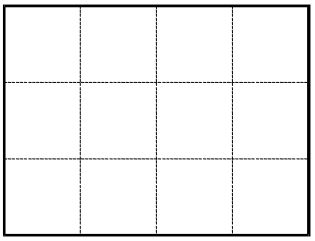
(c)

600	÷	24	=	25
600	÷	.....	=	50

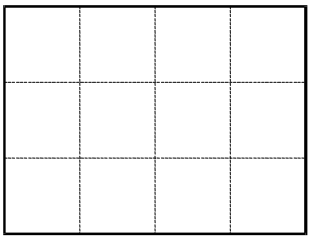
18. You can draw one straight line on the rectangle to make **two triangles**.



(a) Draw one straight line on the rectangle below to make **one square** and **one rectangle** that is not a square.



(b) Draw one straight line on the rectangle below to make **one triangle** and **one quadrilateral**.



19. The number 1 is written on a piece of paper and then passed round the room to each of 50 students in turn. Each student must cross out the number seen and replace it with its double plus 1.

a. Fill in the next three numbers which should appear on the piece of paper.

1 _____ _____ _____

b. The last student does this then announces the final number.

What should the final number be?

_____

END OF EXAM